CAMERON WILDING

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Professional Summary

Electrical and Computer Engineering graduate student specializing in digital hardware design and hardware security. Experienced leading multi-person tape-out teams, developing and verifying RTL in Verilog/SystemVerilog, and applying Python/CUDA tooling to hardware-focused AI research. Seeking full-time hardware design beginning January 2027.

Education

Worcester Polytechnic Institute, Worcester, MA

Master of Science in Electrical and Computer Engineering, **GPA: 4.0/4.0** *May 2025 - Dec 2026*
Relevant Coursework: Computer Architecture, Advanced Digital Design

Bachelor of Science in Electrical and Computer Engineering, **GPA: 3.70/4.0** *Aug 2022 - May 2026*
Relevant Coursework: Advanced Digital Design with FPGAs, Computer Organization, Analog IC Design, CMOS Fundamentals

Publications

Wilding, C., Shaker, M., et al. “Not to Break, but to Attest: Adversarial Probes for Privacy-Preserving LLM Verification”. Under review. Preprint: https://arxiv.org/abs/2608.27954

Work Experience

Gradutate Research Assistant, Worcester Polytechnic Institute *Jan 2026 – Present*

- Conducting research on cryptographic AI governance by developing a system to verify large language model integrity without access to model weights or internals, as part of M.S. thesis
- Analyzed RF emissions from GPUs running AI workloads to extract model information and generated tokens via cache-clock side-channel analysis.

Teaching Assistant, Worcester Polytechnic Institute *Aug 2025 – Jan 2026*

- Supported 31 students in ECE3829: Advanced Digital Design with FPGAs
- Guided development of digital circuits and state machines on the Xilinx Artix-7 in Vivado
- Graded lab work and provided technical support in Verilog

Projects

Power Management IC Design for IoT Applications, WPI Major Qualifying Project *Aug 2025 – May 2026*

- Led digital design for an 8-person team developing a custom ASIC using the Cadence tool suite
- Designed control state machines in Verilog to manage power monitoring and sequencing
- Won second place among all WPI Major Qualifying Projects; taped out May 2026, with testing planned for Fall 2026

Trace Data Compressor, Independent *April 2026*

- Designed a lossless data compressor in SystemVerilog to compress bus data for storage on BRAM
- Achieved a 7.4x compression rate on FPGA, winning Best Rookie Hack at GoatHacks 2026
- Taped out with TinyTapeout in April 2026

Tensor Co-Processor for Croc SoC, Independent *Oct 2025 – Dec 2025*

- Designed systolic array, control logic, and partial product accumulation in SystemVerilog
- Optimized for low area and power, with INT8 inputs accumulating to INT32 for high accuracy
- Verified with UVM in December 2025; taped out April 2026, with testing planned for Fall 2026

CubeSat Support Systems Development, WPI Satellite Development Club *Jan 2024 – Dec 2025*

- Led the Support Systems Subteam of the WPI Satellite Development Club
- Developed a C codebase to control the power board, camera, and data transmission
- Designed the main board in Altium to connect and control the power board and camera over SPI

Technical Skills

Hardware/Software: Vivado, KiCad, LTspice, Altium, Git, Cadence Tool Suite (Virtuoso, Genus, Innovus, Tempus, Spectre), SPICE, Linux, UVM, PyTorch, FPGA, LaTeX

Programming Languages: SystemVerilog, Verilog, C, C++, Python, MATLAB